WADHWANI FOUNDATION

# TourSafe

**Venture Viability Analysis**

SRI SAIRAM ENGINEERING COLLEGE, Chennai

# TourSafe

Smart Real Time Tourist Safety Monitoring & Emergency Response System using Al, Geo-Fencing.

**SRI SAIRAM ENGINEERING COLLEGE, Chennai**

# Venture Team

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**Name: SNEHA C**
Major: BE CSE

**Name: MADHUMITHA S**
Major: BE CSE

**Name: VISHAL L**
Major: BE CSE


# Context

Safety concerns cause drop in tourist arrivals by up to 15%, response time averages 30 - 40 minutes.

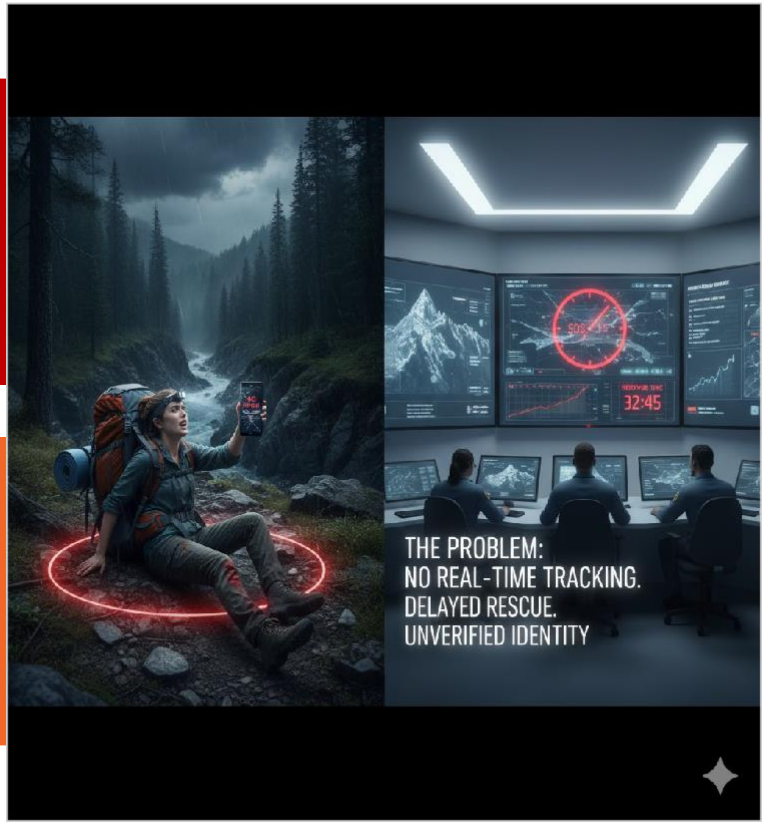
# Problem Statement

## Problem

Tourist safety is compromised by 30-minute emergency delays in remote areas, zero real-time tracking, and slow, unreliable ID verification.

## Impact

Tourists face fatal delays and fear. Authorities battle strained resources and slow ID checks.The industry suffers reputation damage and economic loss



# Problem Statement/Industry

### Problem Being Solved

Tourists face safety issues in remote or high-risk areas, especially in forests, mountains, or politically sensitive regions where network connectivity and traditional monitoring fail. There is No real-time tourist tracking , Slow emergency response, Difficulty in identity verification, Restricted area violations, Lack of predictive safety alerts

### Supporting Data

Safety concerns cause drop in tourist arrivals by up to 15%, response time averages 30 - 40 minutes.

**Source**: UNWTO Report on Tourist Safety and Travel Trends in India, 2024

**Area**
Hospitality

**Industry**
Tourism & -Hospitality

**Domain**
Travel & Tourism

# Problem Analysis

## Affected Stakeholders

Tourists visiting remote or high-risk areas face safety risks due to poor monitoring and delayed emergency response. Local authorities, tourism departments, and law enforcement struggle to track, verify, and assist tourists efficiently. This affects tourist trust, regional tourism revenue, and the overall image of travel safety in India.

## Impact on Stakeholders

Tourist fatalities stem from 30+ min EMS delays in remote areas and no alerts for restricted zones. This risk fuels tourist fear,degrading the travel experience.Authorities battle resource strain and slower rescue due to lack of real-time tracking/manual reporting.The tourism sector faces reputation damage,lost revenue,and higher insurance premiums

## Root Causes

The problem persists due to fragmented, legacy systems for identification (no unified digital ID/KYC). Poor network coverage in high-risk zones constantly cuts off standard GPS tracking. There is a critical lack of AI-based predictive analytics and proactive geo-fencing capability, meaning emergency response remains severely reactive and too slow.

## Personal/Team Connect

We personally felt the fear of isolation during travel. Our passion is deploying AI/TensorFlow skills to move beyond simple tracking and generate instant, life-saving anomaly alerts. We couple this with a Blockchain ID to forge a chain of secure, trusted, real-time help. Our goal: cut emergency response time by 70%, making safety a guaranteed real

# Target Customer Segments

## Primary

All international and domestic tourists traveling to unfamiliar, high-risk or remote regions.

## Secondary

Tourist agencies and government tourism departments.

# Customer Segment & Persona

**Primary Segment**

All international and domestic tourists traveling to unfamiliar, high-risk or remote regions.

**Secondary Segment**

Tourist agencies and government tourism departments.

Persona

**Tourists**

**Age in years: 28**

**Location: Suburban**

**Organizational Role: {Persona's primary role}**

(if applicable)

# Customer Profile

**Education:** College pass out

**Gender:** Female

**Occupation:** Works for a private company

**Interests/Hobbies:** Travel

**Primary Source of Information:** Social Media

**Shopping Preference:** Hybrid

**Comfort with Technology:** Medium

**Favourite Social Media:** Instagram, Twitter, Youtube, Blogs.

**Favourite Offline Gathering Spots:** Tourist spots

# Jobs-to-be-Done

## Functional JTBD

The current system limits tourists from key tasks: they can't travel freely without anxiety or ensure rapid rescue, as help often takes 30+ minutes in remote areas. They struggle to validate identity when papers are lost and can't avoid dangerous zones due to lack of proactive, automated geo-fencing alerts and real-time safety intelligence.

## Emotional JTBD

Tourists seek to feel secure and trusting in a system powered by Blockchain ID, removing the fear of being ignored or unrecognized. They long to feel relieved and worry-free, without the anxiety that a remote mishap could lead to fatal delay. Above all, they wish to feel empowered, with instant control over their safety, not helpless or uncertain.

## Social JTBD

Tourists want to be recognized as responsible travelers, not seen as reckless risk-takers who burden local resources. They seek respect from authorities through verified Blockchain IDs that make their presence trusted.They desire influence by having safety prioritized and being valued as contributors to safe, sustainable tourism,not mere statistics

# Current Alternatives

### Current Alternatives

Tourists currently rely on several manual, reactive, and often insufficient solutions: they purchase expensive travel insurance that relies on slow response; they hire local guides whose job is not real-time monitoring; they carry physical documents for slow verification; and they rely on standard GPS and phones, hoping for signal in remote areas.

### Gaps in Current Alternatives

Current limitations are severe: costly insurance and guides lack real-time tracking, offering only delayed help. Physical documents are easily lost and fail cross-border verification. Tourists feel unsafe as these systems miss the critical time factor—no instant AI alerts or 10-minute rescue guarantee—leaving them anxious and vulnerable.

# Problem Validation (GOOTB)

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## Partial List of Potential Customers/Users Interviewed

**Name:** Riya

**Occupation:** Trekker

**Name:** john

**Occupation:** Hiker

**Name:** Kenzy

**Occupation:** Cave diver

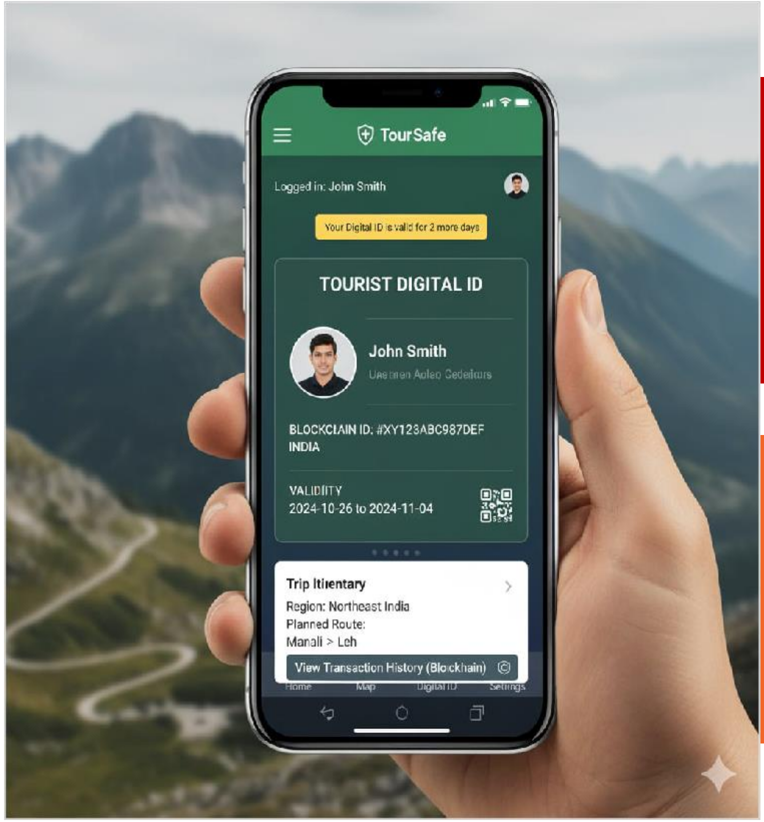
## Problem Validation

**Total customers/users interviewed:**
- In-person: 5
- Virtually: 5

**Total customers/users for whom this problem is important to solve:** 6

**Total customers/users who are dissatisfied with the current alternatives:** 9

# Our Solution




## Solution

TourSafe–Smart Real-Time Tourist Safety Monitoring&Emergency Response System using AI-based Anomaly Detection,Geo-Fencing, and Blockchain-based Digital ID for secure identity and rapid rescue.

## Core Technologies/ Methodologies

Our Tourist Safety app will be build using Mern Stack, flutter, Figma, React.

# Solution Design

### Our Solution

TourSafe–Smart Real-Time Tourist Safety Monitoring&Emergency Response System using AI-based Anomaly Detection,Geo-Fencing, and Blockchain-based Digital ID for secure identity and rapid rescue.

### Key Features

Our Safety app will have Blockchain-based Digital ID,Geo-fencing alerts for restricted zones, SOS emergency features, AI anomaly dashboard for unusual behavior, Authority dashboard for quick response.

### Uniqueness

This solution uniquely combines AI-driven Anomaly Detection with Blockchain-secured Digital ID and real-time Geo-Fencing, delivering predictive safety analytics beyond reactive systems.

**Solution Format:**
Digital Product - TourSafe : Tourist Safety App

**Core Technologies/ Methodologies:**
Our Tourist Safety app will be build using Mern Stack, flutter, Figma, React.

# Solution Benefits

### Functional Benefits

Ensure personal safety through real-time tracking,get instant SOS support,access verified digital ID,receive early risk alerts, and travel confidently with AI-driven protection and secure data privacy

### Emotional Benefits

Users gain peace of mind with continuous monitoring and protection, easing anxiety about remote accidents and empowering them with immediate control, turning helplessness into confidence.

### Social Benefits

Gain trust and admiration for choosing safe, smart travel, inspire others to explore confidently, earn respect for responsible tourism, and enjoy peace of mind while sharing secure and positive travel

### Macro Benefits

TourSafe improves tourist safety through real-time alerts, reducing accidents and saving lives. It strengthens tourism’s contribution to the economy, promotes data security and responsible travel.

# Competitors

## Direct

HollieGuard, TravelSafe
Everbridge, International SOS

## Indirect

Garmin inReach, Spot, Life360
Google Maps (offline), Travel Insurance providers (e.g., Allianz Travel), Airbnb

## Our UVP

TourSafe uses AI and blockchain for instant, verified rescue, cutting response times globally.

# Competitors

### Direct Competitors

HollieGuard, TravelSafe

### Indirect Competitors

Garmin inReach, Spot, Life360

### Direct Competitors Globally

Everbridge, International SOS

### Indirect Competitors Globally

Google Maps (offline), Travel Insurance providers (e.g., Allianz Travel), Airbnb

# Macro Analysis

## Favourable Trends

| AREA | DESCRIPTION |
|---|---|
| **Technology** | AI and IoT adoption enables real-time tourist tracking and predictive safety alerts. |
| **Economy** | Rising travel spending and tourism recovery boost demand for safe, convenient travel apps. |

## Unfavourable Trends

| AREA | DESCRIPTION |
|---|---|
| **Climate/Environment** | Dense forests,deep valleys,or extreme weather conditions may interfere with GPS and mobile signals |
| **Legal** | Authorities relying solely on the system may overlook the need for manual help. |

### Data Sources:

Information is based on UNWTO reports, government tourism guidelines, AI & blockchain research, tech docs (Google Maps, Firebase), and news on tourist safety incidents.

# Back-of-the-Envelope Financial Projections

**Currency:** Indian Rupee (INR)

**Chosen Business Model:** Service Provider

| AREA | YEAR 1 | YEAR 2 | YEAR 3 |
|---|---|---|---|
| **Revenues** | 100000000 | 275000000 | 630000000 |
| **Total Expenses** | 72700000 | 104140000 | 143232000 |
| **Profit** | 27300000 | 170860000 | 486768000 |

# Prototype

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## Prototype Format

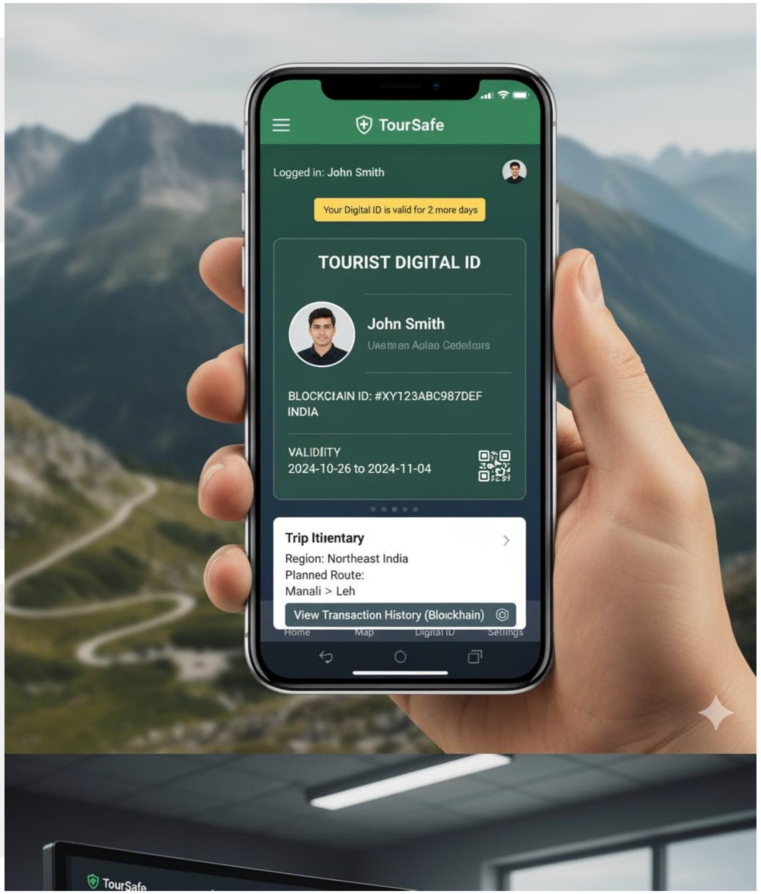
The prototype is a Digital Hybrid combining a Figma Mockup with a Gen AI Demo showcasing UI/UX

## Functionality included in the Prototype

The TourSafe prototype features: Blockchain ID view, AI Risk Score panel, One-Touch SOS, Shake-to-Alert, Anomaly Simulation, and Contextual Rescue Guidance.

## Functionality NOT included in the Prototype

The prototype will exclude payment gateway integration, full end-to-end blockchain writing/reading, advanced AI model training/retraining, and live database integration with authorities.



# Prototype Validation

| Number of users engaged with? | How many people liked or loved the prototype? | How many people were either neutral or mostly unhappy with the prototype? |
|---|---|---|
| 18 | 14 | 4 |

## Prototype Feedback

**What aspects of the prototype did the users LOVE?**

Users loved the Proactive AI Anomaly Detection (predicting falls/incapacitation before SOS. Authorities praised the Instant Blockchain Digital ID Verification and the text Gen AI's Contextual Rescue Guidance (cutting response time). Tourists found the UI and "Shake-to-Alert" feature simple and highly reassuring.

**What aspects of the prototype were DISLIKED by the users?**

Users had reservations about two areas: the initial mandatory KYC process for the Blockchain ID, perceiving it as friction slash delay. They were also neutral slash disliked the AI Anomaly Risk Score displayed in the app, finding the numerical value confusing unless it was explicitly high slash low.

# Competition Analysis

| COMPETITOR NAME | TYPE | STRENGTHS | WEAKNESSES |
|---|---|---|---|
| International SOS | Direct | High global trust/reputation | No predictive AI Anomaly Detection |
| Garmin inReach | Indirect | satellite connectivity,works on all | No AI monitoring, GeoFencing alerts |
| GeoSure | Direct | Real time Risk Scores, Visual Data | offers zero emergency response |
| bSafe | Direct | Focus on women's safety | Reactive SOS only |

**Our Product/Service will be better than the competitors' solutions because:**

TourSafe is better because it offers: 1. Predictive AI Anomaly Detection; 2. Secure Blockchain Digital ID; 3. Unified app/dashboard for a guaranteed 70% faster rescue time.

# Market Size & GTM

GTM Channels

### Digital

LinkedIn, SEO, Targeted PPC Ads (Google/LinkedIn), Email Marketing (for thought leadership/case studies), Webinars (for live demoing the AI dashboard).

### Physical

TRM conferences, GovTech trade shows (like WTM and IITF), direct consultations through B2G demos of the AI dashboard, and sponsorships at major high-impact events.

**Market Size**

**Globally:**
**Revised TAM Estimate:Global TRM Market:₹10Trillion**

**Growth Rate:**
**Annual Growth Rate: 8.1% to 8.8% CAGR**

Source: Allied Market Research, Fortune Business Insights.

# Market Size

## Total Addressable Market (TAM)
Revised TAM Estimate:Global TRM Market:₹10Trillion

## Serviceable Available Market (SAM)
₹2.6 trillion (USD 32 billion).

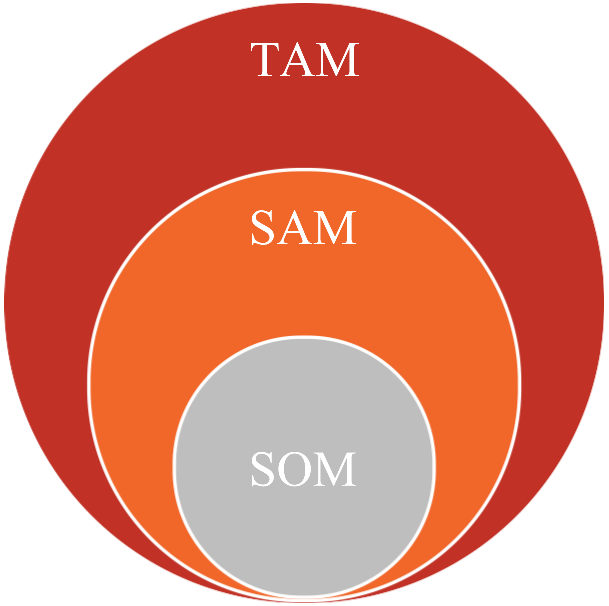
## Serviceable Obtainable Market (SOM)
₹37.5 crore in 2 years.

## Assumptions

Our SAM calculation assumes we first target government and authority clients in five high-risk tourism regions such as the Alps and the Himalayas. It is based on 25 million adventure tourists who are willing to pay for AI and blockchain-based safety ,
20000000,
16

## Sources of Research

Allied Market Research, Fortune Business Insights.



# Revenue Models / Pricing

**Revenue Model (Primary)**

Subscription Model

**Revenue Model (Secondary)**

Saas (Software as a Service) Model

# Lean Canvas

### Problem
Tourist safety is compromised by 30-minute emergency delays in remote areas, zero real-time tracking, and slow, unreliable ID verification.

### Solution
TourSafe–Smart Real-Time Tourist Safety Monitoring&Emergency Response System using AI-based Anomaly Detection,Geo-Fencing, and Blockchain-based Digital ID for secure identity and rapid rescue.

### Key Metric
Key metrics are Average Rescue Time Reduction, Monthly Recurring Revenue (MRR), Subscription Renewal Rate, and the False Positive Alert Rate to ensure AI accuracy and system trust.

### Unique value Proposition
TourSafe uses AI and blockchain for instant, verified rescue, cutting response times globally.

### Unfair Advantage
Proprietary **AI anomaly detection** and **Blockchain-Secured IDs** are hard-to-copy, ensuring instant, verified rescue.

### Channels
Primary (App Stores): Essential for direct, instant delivery to solo tourists. Secondary (B2B/Partnerships): Crucial for mass adoption via Agencies and Corporate Managers.

### Customer Segment
All international and domestic tourists traveling to unfamiliar, high-risk or remote regions.,Tourist agencies and government tourism departments.

### Cost Structure
The main costs are Cloud Infrastructure (for real-time AI/tracking), Platform Development (AI, app updates, blockchain security), and 24/7 Multilingual Emergency Support staff and operations.

### Revenue Streams
Subscription Model, Saas (Software as a Service) Model

# Go-to-Market Approach

**Geographic Focus**

Targeting Himachal Pradesh/Kerala state governments (India) for an initial B2G pilot contract in a high-risk trekking zone. Building visibility at global TRM conferences for Year 2 launch.

**Digital Marketing Channels**

LinkedIn, SEO, Targeted PPC Ads (Google/LinkedIn), Email Marketing (for thought leadership/case studies), Webinars (for live demoing the AI dashboard).

**PRIMARY CUSTOMER SEGMENT**

All international and domestic tourists traveling to unfamiliar, high-risk or remote regions.

**Marketing KPIs**

Contract Win Rate, LTV:CAC Ratio, Net Revenue Retention (NRR) from existing B2G contracts, and End-User Adoption Rate (percentage of tourists using the app).

**Physical Marketing Channels**

TRM conferences, GovTech trade shows (like WTM and IITF), direct consultations through B2G demos of the AI dashboard, and sponsorships at major high-impact events.

**UVP**

TourSafe uses AI and blockchain for instant, verified rescue, cutting response times globally.

**Competitors' GTM**

Intl SOS: direct sales, consulting, webinars, and TRM conferences. Garmin: influencer marketing, user-generated content (customer stories), retail distribution, and targeted social media ads.

**GTM Partners**

TRM Consultancies (e.g., Control Risks) for leads/credibility. Blockchain Platforms (Hyperledger/IBM) for joint bids/tech content. Local Mountain Rescue/Tourist Police for case studies/endorsement.

# Sales & Customer Service

**Customer Service**

AI chatbots and multilingual 24/7 voice support for tourists. Dedicated 24/7 phone support, on-site training, and account management for B2G authority clients.

**Distribution Channels**

Primary (App Stores): Essential for direct, instant delivery to solo tourists. Secondary (B2B/Partnerships): Crucial for mass adoption via Agencies and Corporate Managers.

**Digital Sales Channels**

Government e-Procurement Portals (GeM, GePNIC, TendersInfo), Direct RFP Submission (Digital), and Specialized B2G SEO/Content Marketing.

**PRIMARY CUSTOMER SEGMENT**

All international and domestic tourists traveling to unfamiliar, high-risk or remote regions.

**UVP:**

TourSafe uses AI and blockchain for instant, verified rescue, cutting response times globally.

**Physical Sales Channels**

Direct Field Sales (B2G Consultations), On-Site AI Dashboard Demos (Pilot Programs), GovTech Conferences (India AI Impact Summit), and Major Tourism Marts (SATTE/ITM) for ministerial access.

**Sales KPIs**

Win Rate (percentage of bids won), Average Sales Cycle Length, LTV:CAC Ratio, and Annual Recurring Revenue (ARR) for tracking the high-value B2G contract pipeline efficiency.

https://crowdsourced-civic-lssue-reporting-pi.vercel.app/

**GTM Partners**

Digital: NPS (end-user loyalty) and CSAT (post-SOS incident). Authority: formal CPARS reviews evaluating dashboard usefulness and the success rate of AI guidance.

https://drive.google.com/drive/folders/1zU6hL3iySTzmR2vWMv7j17vR3g7oqt0C

# Financials

## Revenue Models/Streams

- Subscription Model
- Saas (Software as a Service) Model

## Pricing

- **Unit of Sale:** SUBSCRIPTION
- **Selling price per unit:** ₹999

## First Year Projections

Revenues: ₹54,43,751

Operating Profits: ₹10,43,59

# Revenue Models / Pricing

**Revenue Model (Primary)**

Subscription Model

| Unit of Sale | Sale Price per Unit | Expected units to be sold in Year 1 | Expected growth in monthly sales |
|---|---|---|---|
| SUBSCRIPTION | ₹999 | 250 | 10% |

# Costs & Revenues: Key Assumptions

[object Object]

# Profit & Loss Projections: Summary

[object Object]

# Performance & Break-Even Analysis

| | |
|---|---|
| **Year 1 Revenues**<br>₹54,43,751 | **Gross Profits for Year 1**<br>₹47,16,401 |
| **Net Profits for Year 1**<br>₹10,43,59 | **Break-even Month**<br>month 10 |

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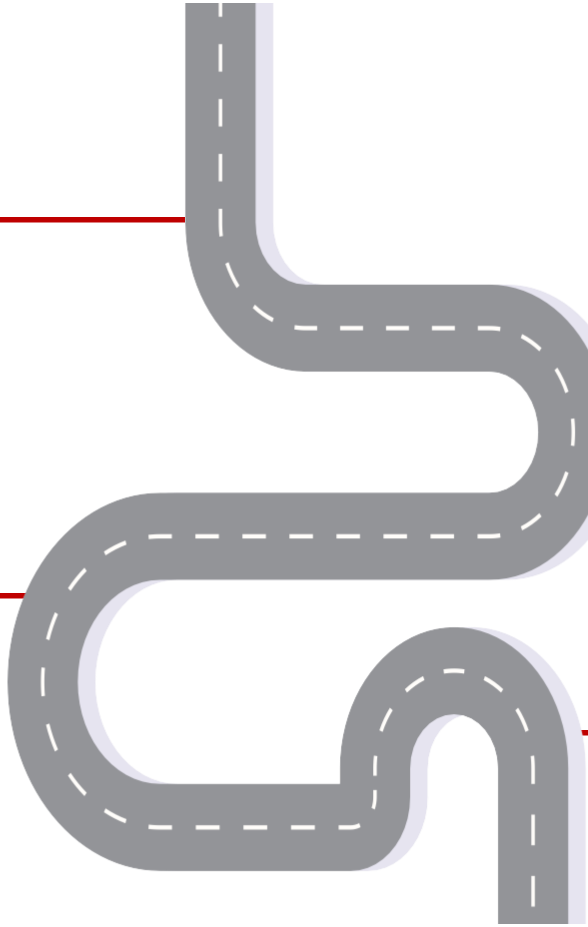
# Next Steps

**Goals for Months 10-12**

Deploy field pilot, optimize AI, finalize biz mode

**Goals for Months 4-6**

Finalize AI/Blockchain integration &security audit



**Goals for Months 7-9**

Secure B2G pilot MoU & develop commercial proposal

**Goals for Months 1-3**

Develop MERN MVP, AI model train, launch alpha

# Venture Viability Assessment

## Venture Viability Index

88.88%

### Strengths

TourSafe's key strengths are its differentiated value proposition (5/5) and mission commitment (5/5). The AI/TensorFlow and Blockchain Digital ID create a unique, proactive safety ecosystem, eliminating reactive SOS delays. The focus is on a critical, high-growth market need (reducing EMS time by 70%) with a highly dedicated team.

### Areas of Improvement

The key area for improvement is execution risk (3/5). We need to de-risk the B2G sales model and financial projections. The slow government acquisition cycle impacts revenue (3/5). We must secure specialized B2G sales expertise and contingency funding to manage the Year 1 loss and the complexity of integrating with diverse global EMS systems.

# Next Steps

| TIMELINE | GOALS | TEAM NEEDED | PHYSICAL RESOURCES NEEDED | FUNDS NEEDED |
| --- | --- | --- | --- | --- |
| **Months 1-3** | Develop MERN MVP, AI model train, launch alpha | 3 | GPU Server | ₹5 Lac |
| **Months 4-6** | Finalize AI/Blockchain integration &security audit | 3 | Cloud Acce | ₹4 Lac |
| **Months 7-9** | Secure B2G pilot MoU & develop commercial proposal | 2 | Co-working | ₹6 Lac |
| **Months 10-12** | Deploy field pilot, optimize AI, finalize biz mode | 2 | Test Geo-A | ₹5 Lac |

# Venture Team

**Name: SNEHA C**

**University/College:** SRI SAIRAM ENGINEERING COLLEGE

**Major:**BE CSE

**Key Skills:** Coding,UX design,Web dev

**Role in the Venture:** Chief Technologist

Keen on continuing with the venture:

**Yes**

**Name: MADHUMITHA S**

**University/College:** SRI SAIRAM ENGINEERING COLLEGE

**Major:**BE CSE

**Key Skills:** Presenting, UX, Coding

**Role in the Venture:** Chief Strategist

Keen on continuing with the venture:

**Yes**

**Name: VISHAL L**

**University/College:** SRI SAIRAM ENGINEERING COLLEGE

**Major:**BE CSE

**Key Skills:** UX design,coding,web dev

**Role in the Venture:** Chief Product Office

Keen on continuing with the venture:

**Yes**

**Current Mentors:**
College Guide + B2G Sales Expert + AI/Blockchain Architect & EMS Consultant

**Mentors Needed in these Areas:**
B2Gsales strategy,pilot deployment logistics,AI optimization,and fundraise

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# Venture Summary

## OVERVIEW

We at TourSafe are building an intelligent,technology-driven solution dedicated to redefining global tourist safety.We merge AI-based predictive anomaly detection,Geo-Fencing,and Blockchain Digital ID into a unified system to tackle inadequate monitoring and slow emergency response.Currently,we are developing the mobile app and authority dashboard prototype, focused on dramatically reducing emergency response times by 70% and improving coordination.We are committed to fostering traveler secure.

## Mission

Our mission is to redefine tourist safety by seamlessly merging AI intelligence, geo-awareness, and Blockchain-secured trust, guaranteeing a proactive, real-time safety net that cuts emergency response time by $\mathbf{70\%}$, fostering global traveler confidence, and promoting safe, technology-enabled tourism aligned with UN SDGs.

## Social/Economic Relevance

As a society, we must ensure safe, responsible tourism. When visitors face preventable risks due to slow response or unverified identity, it erodes trust, damages economies, and fails humanitarian standards. TourSafe supports SDG 8 (Economic Growth) and SDG 16 (Peace/Justice) by making travel inherently safer and more trustworthy for all.

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# Thank You!

Our mission is to redefine tourist safety by seamlessly merging AI intelligence, geo-awareness, and Blockchain-secured trust, guaranteeing a proactive, real-time safety net that cuts emergency response time by $\mathbf{70\%}$, fostering global traveler confidence, and promoting safe, technology-enabled tourism aligned with UN SDGs.